

BVRIT HYDERABAD College Of Engineering For Women

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Aim: Explore the features of ES6 like arrow functions, callbacks, promises, async/await. Implement an application for reading the weather information from openweathermap.org and display the information in the form of a graph on the web page.

```
const API_KEY = "17a7e1cc3e0a9e0f9304bd814e97e23a";

const cityInput = document.getElementById("cityInput");

const getWeatherBtn = document.getElementById("getWeatherBtn");

const messageBox = document.getElementById("message");

let weatherChart = null; // will hold our chart, so we can redraw it later

getWeatherBtn.addEventListener("click", () => {

  const city = cityInput.value.trim();

  if (city === "") {

    messageBox.textContent = "Please type a city name first.";

    return;

  }

  messageBox.textContent = "Loading...";

  getWeatherData(city); // call our async function below

});

async function getWeatherData(city) {

  try {

    const url =

`https://api.openweathermap.org/data/2.5/forecast?q=\${city}&units=metric&appid=\${API\_KEY}

    ;

    const response = await fetch(url);
```

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```
const data = await response.json();

if (data.cod !== "200") {

    messageBox.textContent = "City not found. Please check the spelling.";

    return;

}

messageBox.textContent = ""; // clear any old message

    const times = data.list.map((entry) => entry.dt_txt.slice(5, 16)); // e.g. "08-15 09:00"

const temps = data.list.map((entry) => entry.main.temp);

drawGraph(times, temps, city);

} catch (error) {

    console.log("Something went wrong:", error);

    messageBox.textContent = "Something went wrong. Check your internet or API key.";

}

}

function drawGraph(labels, temperatures, city) {

    const ctx = document.getElementById("weatherChart").getContext("2d");

    if (weatherChart !== null) {

        weatherChart.destroy();

    }

    weatherChart = new Chart(ctx, {

        type: "line",

        data: {

            labels: labels, // X axis: date/time
```

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```
datasets: [  
  {  
    label: `Temperature in ${city} (°C)`,  
    data: temperatures, // Y axis: temperature values  
    borderColor: "blue",  
    fill: false,  
    tension: 0.2,  
  },  
],  
},  
options: {  
  responsive: true,  
  scales: {  
    x: { title: { display: true, text: "Date / Time" } },  
    y: { title: { display: true, text: "Temperature (°C)" } },  
  },  
},  
});  
}
```

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```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8" />

  <title>ES6 Weather Graph App</title>

  <!-- Chart.js library: lets us draw a graph very easily, no need to
build one ourselves -->

  <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>

  <style>

    body {

      font-family: Arial, sans-serif;

      max-width: 700px;

      margin: 40px auto;

      text-align: center;

    }

    input, button {

      padding: 8px;

      font-size: 16px;

    }

    #message {

      color: red;

      margin-top: 10px;

    }

  </style>
```

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</head>

<body>

<h1>🌤 Weather Forecast Graph</h1>

<input type="text" id="cityInput" placeholder="Enter city name" />

<button id="getWeatherBtn">Get Weather</button>

<p id="message"></p>

<canvas id="weatherChart" width="600" height="300"></canvas>

<script src="app.js"></script>

</body>

</html>

file:///D:/Chrome%20Downloads/nodejs/index.html/26wh5a0508

🌤 Weather Forecast Graph

Chennai

Get Weather

